THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 38	Perimeter $= 2(l + b)$.
2	**(a)** 8	The number below the line.
3	**(a)** 8	Diameter $= 2\times$ radius.
4	**(c)** 4	A regular polygon with 4 sides has 4 lines of symmetry.
5	**(d)** -7	The opposite has the same distance from 0 on the other side.
6	**(c)** 24	Perimeter $= 4\times$ side.
7	**(c)** $\frac{5}{7}$	Add the numerators; keep the denominator.
8	**(d)** 5	Radius $=$ diameter $\div 2$.
9	**(a)** 1	Letter A.
10	**(a)** -4	Add 1.
11	**(b)** 54	Area $= l \times b$.
12	**(a)** $\frac{5}{11}$	Subtract the numerators; keep the denominator.
13	**(b)** a compass	Property
14	**(c)** F	It cannot be folded into two matching halves.
15	**(a)** -4	Subtract 1.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Perimeter $= 2(l + b)$ (1); $2(17 + 9) = 52$ m (1).	1+1
17	Denominators are equal (1); $\frac{5}{9}$ (1).	1+1
18	Diameter $= 2 \times 6 = 12$ cm (1); Longest chord is the diameter, 12 cm (1).	1+1
19	A regular polygon has as many lines as sides (1); 4 lines (1).	1+1
20	Opposite of 5 is -5 (1); Opposite of -5 is 5 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Perimeter $= 4a = 4 \times 22 = 88$ m (1); Area $= a^2 = 22^2 = 484$ m^2 (1); State units (1).	1+1+1
22	Multiply numerator and denominator by 2, 3, 4 (1); $2/6$, $3/9$, $4/12$ (2).	1+2
23	Radius $= 5$ cm (1); Chord 7 cm vs diameter 10 cm (1); No: the diameter is the longest chord (1).	1+1+1
24	Lines $= 4$ (1); Order $= 4$ (1); Smallest angle $= 90^\circ$ (1).	1+1+1
	OR: Subtracting an integer is the same as adding its opposite (1); $5 + (-8)$ (1); $= -3$ (1).	1+1+1
25	Perimeter $= 2(60 + 40) = 200$ m (1); Wire needed 200 m (1); Area $= 60 \times 40 = 2400$ m^2 (1).	1+1+1
	OR: Equal denominators: compare numerators (1); $\frac{5}{11} < \frac{8}{11}$ (1); Greater: $\frac{8}{11}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Area of sheet $= 30 \times 20 = 600$ cm^2 (1); Area of square $= 8^2 = 64$ cm^2 (1); Area left $= 600 - 64 = 536$ cm^2 (1); Units: cm^2 (1); State the answer (1).	1+1+1+1+1
	OR: Used =\frac{2}{5}$\times 20 = 8$ m (1); Left $= 20 - 8 = 12$ m (1); Used + left = total (1); State the units (2).	1+1+1+2
27	Draw AB = 6 cm (1); Draw perpendicular at B and mark BC = 4 cm (1); Draw perpendicular at A and mark AD = 4 cm, join D and C (1); Check that DC = 6 cm (1); Perimeter $= 2(6 + 4) = 20$ cm (1).	1+1+1+1+1
	OR: Lines of symmetry $= 3$ (1); Order $= 3$ (1); Angle $= 360 \div 3 = 120^\circ$ (1); The polygon is a 3-sided regular polygon (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 49	Area $= a \times a$.
2	**(c)** 8	Multiply the numerator by 4 too.
3	**(a)** 9	Opposite sides of a rectangle are equal.
4	**(c)** 180	Angle $= 360^\circ \div$ order.
5	**(d)** 5	Use the number line or tokens.
6	**(a)** 15	All three sides are equal.
7	**(c)** 36	Divide 48 by 4 and multiply by 3.
8	**(d)** 90	All angles of a square are right angles.
9	**(a)** infinitely many lines of symmetry	Symmetry property
10	**(b)** 2	The number to the right on the number line is greater.
11	**(a)** 24	Six equal sides.
12	**(c)** $\frac{7}{3}$	2×3+1 = 7, so the fraction is 7 over 3.
13	**(c)** 360	4 right angles.
14	**(d)** 3	Order = number of times it fits.
15	**(c)** 6	Distance is never negative.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area $= l \times b$ (1); $14 \times 9 = 126$ cm^2 (1).	1+1
17	$17 \div 5$ gives quotient 3, remainder 2 (1); $3\frac{2}{5}$ (1).	1+1
18	A rectangle (1); Other sides are 8 cm and 5 cm (1).	1+1
19	Angle $= 360^\circ \div 4$ (1); 90° (1).	1+1
20	Successor $= -3$ (1); Predecessor $= -5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Open the compass to 3 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 3 cm from O (1).	1+1+1
22	Imagine folding the letter (1); A: yes; H: yes; B: yes; C: yes (1); Check each fold (1).	1+1+1
23	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -7, -2, 0, 3, 5 (1); Further from 0 on the negative side means smaller (1).	1+1+1
24	Side $= 36 \div 4 = 9$ cm (1); Area $= 9^2 = 81$ cm^2 (1); State units (1).	1+1+1
	OR: 36 divided by 3 is 12 (1); Multiply by 2: $12 \times 2 = 24$ (1); State the answer (1).	1+1+1
25	Perimeter $= 2(9 + 4) = 26$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1
	OR: A mirror image is at equal distance on the other side (1); The image is 2 units to the right of the mirror (1); Distance between point and image $= 2 \times 2 = 4$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	After going down: $2 - 5 = -3$ (1); After going up: $-3 + 1 = -2$ (1); Final floor -2 (1); Below ground (2).	1+1+1+2
	OR: Side $= 10$ m (1); Rectangle breadth: $2(12 + b) = 40$ gives $b = 8$ m (1); Square area $= 100$ m^2 (1); Rectangle area $= 96$ m^2 (1); The square has the larger area (1).	1+1+1+1+1
27	Add the numerators: $3 + 4 + 5 = 12$ (1); Sum $= \frac{12}{7}$ (1); As a mixed number: $1\frac{5}{7}$ (1); Reduce if possible (2).	1+1+1+2

	OR: Draw AB = 4 cm (1); Raise perpendiculars at A and B of length 4 cm (1); Join the ends to complete the square (1); Perimeter $= 16$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6

Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	Half the perimeter minus the length.
2	**(c)** 9	The number below the line.
3	**(a)** 14	Diameter $= 2\times$ radius.
4	**(d)** 3	A regular polygon with 3 sides has 3 lines of symmetry.
5	**(d)** negative integers	Definition
6	**(c)** 48	Perimeter $= 2(l + b)$.
7	**(c)** $\frac{9}{11}$	Add the numerators; keep the denominator.
8	**(c)** 8	Radius $=$ diameter $\div 2$.
9	**(b)** 2	Letter H.
10	**(b)** 9	The opposite has the same distance from 0 on the other side.
11	**(a)** 36	Perimeter $= 4\times$ side.
12	**(a)** $\frac{6}{13}$	Subtract the numerators; keep the denominator.
13	**(d)** chord	Property
14	**(b)** G	It cannot be folded into two matching halves.
15	**(a)** 1	Add 1.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Perimeter $= 4 \times 13 = 52$ cm (1); Area $= 13 \times 13 = 169$ cm^2 (1).	1+1
17	HCF of 12 and 18 is 6 (1); $\frac{2}{3}$ (1).	1+1
18	All sides are 7 cm (1); Perimeter $= 4 \times 7 = 28$ cm (1).	1+1
19	Fold along a vertical line (1); A, H, T (1).	1+1
20	Use the number line (1); $= -3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	New temperature $= -3 + 8$ (1); 5°C (1); State units (1).	1+1+1
22	Perimeter $= 4a = 4 \times 30 = 120$ m (1); Area $= a^2 = 30^2 = 900$ m^2 (1); State units (1).	1+1+1
23	Multiply numerator and denominator by 2, 3, 4 (1); $4/10$, $6/15$, $8/20$ (2).	1+2
24	Radius $= 8$ cm (1); Chord 15 cm vs diameter 16 cm (1); No: the diameter is the longest chord (1).	1+1+1
	OR: Lines $= 3$ (1); Order $= 3$ (1); Smallest angle $= 120^\circ$ (1).	1+1+1
25	Subtracting an integer is the same as adding its opposite (1); $-3 + (-4)$ (1); $= -7$ (1).	1+1+1
	OR: Perimeter $= 2(75 + 50) = 250$ m (1); Wire needed 250 m (1); Area $= 75 \times 50 = 3750$ m^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Check both folds (1); (a) H, O, X (1); (b) H, O, X, Z, N (1); Order 2 means it fits after a half turn (2).	1+1+1+2
	OR: Descending: 6, 3, -1, -4 (1); Smallest -4, largest 6 (1); Sum $= 2$ (1); Mark each at its distance from 0 (2).	1+1+1+2
27	Area of sheet $= 40 \times 25 = 1000$ cm^2 (1); Area of square $= 10^2 = 100$ cm^2 (1); Area left $= 1000 - 100 = 900$ cm^2 (1); Units: cm^2 (1); State the answer (1).	1+1+1+1+1
	OR: Used $=\frac{3}{8}\times 40 = 15$ m (1); Left $= 40 - 15 = 25$ m (1); Used + left = total (1); State the units (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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